

Lab 08: Motorbike Braking System

16/12/2025

Abstract

This report evaluates the braking limits and hydraulic component sizing for a high-performance motorcycle.

- **Deceleration Limits:** At low speeds, braking is restricted by road adhesion (0.7g) rather than the 1.14g geometric flipping limit.
- **Aerodynamic Benefit:** At racing speeds (315 km/h), aerodynamic drag significantly increases stability, pushing the flipping limit to 1.63g.
- **Hardware Efficiency:** The specified 36/38 mm calipers and 19x18 master cylinder allow the rider to reach maximum deceleration with 68.6 N of force, which is well within professional ergonomic limits.

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1. Introduction

The objective of this section is to design and analyze the braking performance of a high-performance motorcycle.

The analysis focuses on determining the physical limits of deceleration (flipping vs. adherence) and sizing the hydraulic components required to achieve these limits. The vehicle parameters used for all calculations are summarized here:

- Wheelbase: 1435 mm
- CoG distance from rear axle: 665 mm
- CoG height: 677 mm
- Bike Mass: 157 kg
- Rider Mass: 70 kg
- Total Mass: 227 kg
- Road Friction Coefficient: 0.7

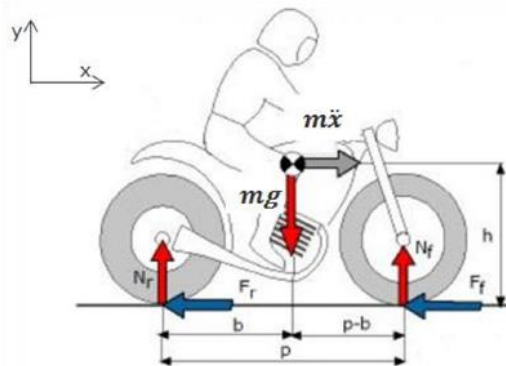


Figure 1

2. Longitudinal dynamics

2.1 Deceleration limits

- Static load distribution
 - o 46/54
 - Computed from the longitudinal moment equilibrium
- Deceleration at flipping limit, without considering aerodynamic effects:
 - o 1.14g
 - The flipping limit occurs when the rear wheel loses contact with the ground, neglecting aerodynamic effects, this limit is purely geometry-dependent
- Without considering aerodynamic effects, the maximum deceleration achievable if $\mu_{tire_road} = 0.7$ is:
 - o 0.7g
 - Computed from longitudinal force and moment equilibrium
- Brake repartition front-rear needed to achieve this deceleration:
 - o 79-21
 - Longitudinal force distribution front and rear is computed for the given acceleration and longitudinal load transfer

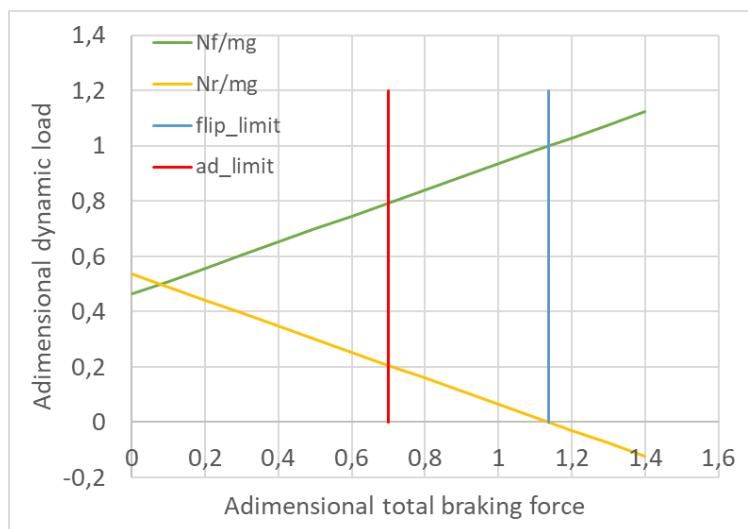


Figure 2

From the figure it is possible to recognize:

- Adimensional braking force, related to acceleration
 - o Flipping limit
 - o Adhesion limit

- Front and rear force distribution

2.2 Front brake only

- If rider decides to use only front brake, the deceleration achieved is:
 - 0.48g
 - The equilibrium is computed again, but with front force only

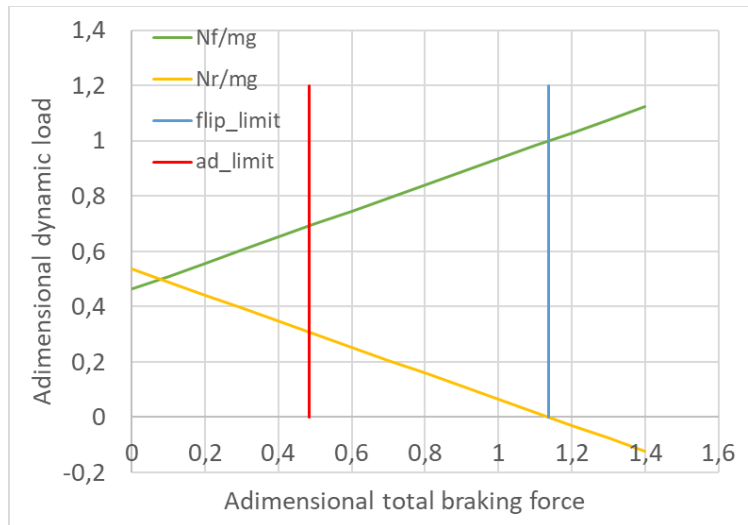


Figure 3

From the figure it is possible to recognize:

- Adimensional braking force, related to acceleration
 - Flipping limit
 - Adhesion limit

2.3 Rear brake only

- If rider decides to use only rear brake, the deceleration achieved is:
 - 0.28g
 - The equilibrium is computed again, but with rear force only

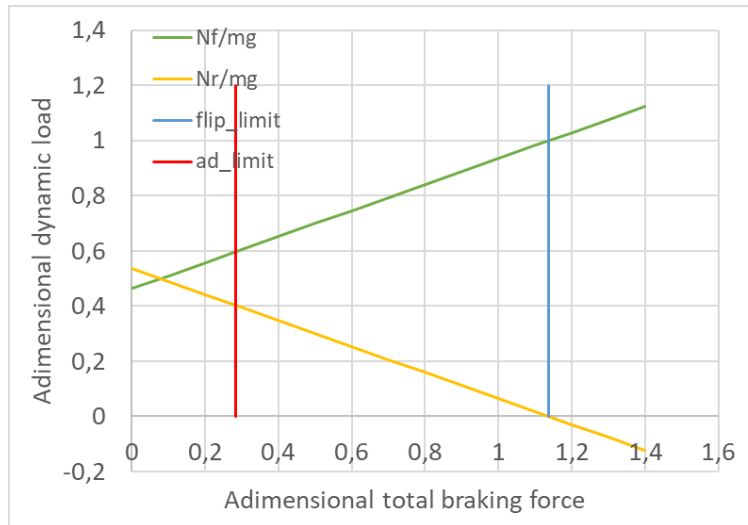


Figure 4

From the figure it is possible to recognize:

- Adimensional braking force, related to acceleration
 - o Flipping limit
 - o Adhesion limit

2.4 Deceleration with aerodynamic effects

- Considering aerodynamic effects, the maximum acceleration achievable at flipping limit considering following conditions ($v = 315 \text{ km/h}$; $C_d = 0.6$; $A = 0.4 \text{ m}^2$; $\rho = 1,2 \text{ kg/m}^3$) is:
 - o 1.63g
 - Effects of drag
 - Helps to stop the motorcycle
 - Different load transfer distribution due to application of aerodynamic force approximated at the CoG

3. Braking system

The front braking system consists of two 340 mm discs and 36/38 mm calipers (M4 configuration). and we consider $\mu_{\text{disk} - \text{pad}} = 0.5$ (disc/pad friction coefficient).

- Pressure in the system at flipping limit:
 - 11.4 bar
- Which is rider force at brake lever if using 19X18 master cylinder
 - 68.6 N
 - Consider rider finger at 100mm from M/C fulcrum.
 - Consider also internal spring preload contribution.
 - Consider +10% due to different inefficiencies like internal components friction.
- Using 19x20 master cylinder
 - 76.2 N

4. Conclusion

The analysis demonstrates that at low speeds, the motorcycle is adherence-limited while at racing speeds, aerodynamics significantly increases stability against flipping. The sizing of the 36/38 caliper with a 19x18 M/C ensures that the flipping limit can be reached with a hand force well within the ergonomic limits of a professional rider.